INCH-POUND

MIL-PRF-19500/211B
AMENDMENT 5
12 February 1999
SUPERSEDING
AMENDMENT 4
2 March 1992

PERFORMANCE SPECIFICATION

SEMICONDUCTOR DEVICE, DIODE, SILICON, POWER RECTIFER,
TYPES 1N3164, 1N3168, 1N3170, 1N3172, 1N3174, 1N3175,
1N3176, 1N3177, AND R TYPES, JANTX AND JANTXV

This amendment forms a part of MIL-PRF-19500/211B, dated 18 May 1983, and is approved for all Departments and Agencies of the Department of Defense.

PAGE 1

Title: Delete and substitute as printed above.

1.5: Delete $R_{\theta JX}$ = 0.15°C/W and substitute: $R_{\theta JX}$ = 0.20°C/W .

PAGE 2

Add 3.3.1 after 3.3:

3.3.1 Processing exemptions. Suppliers to this detail specification are exempt from the following restrictions stated in MIL-PRF-19500 regarding offshore wafer processing:

a. Beveling operation.

b. Isotropic etching of die surface.

c. Die attach operation which promotes any diffusion of metal alloy into silicon die.

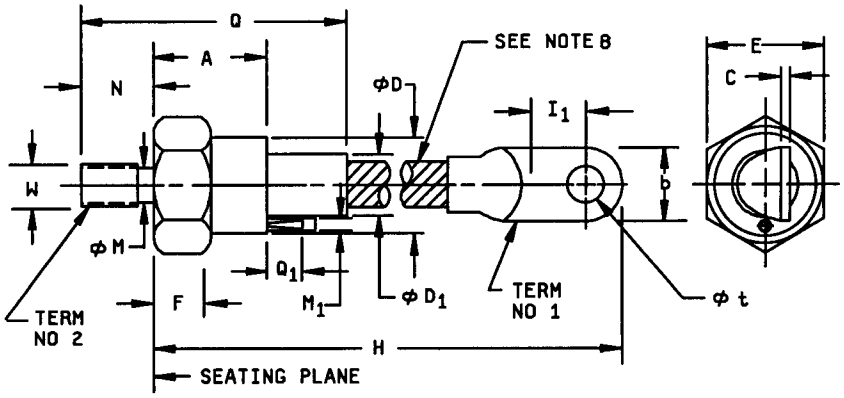
Add 4.2.1 after 4.2:

4.2.1 Qualification exemption. Suppliers to this detail specification are exempt from the basic plant retention of qualification clause of MIL-PRF-19500 appendix E sections 40.3 and 10.2 only if the device manufacturer has previously qualified this device at least once at the basic plant location.

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* FIGURE 1: Delete and substitute the new figure 1 as printed below:



NOTES:
1. Metric equivalents are given for general information only.
2. Complete threads to extend to within 2.5 threads of seating plane.
3. .750-16 UNF-2A. Maximum pitch diameter of plated threads shall be basic pitch diameter. .7094 (18.019 mm) ref. (Screw Thread Standards for Federal Services) FED-STD-H28.
4. Angular orientation of terminal and tabulation with respect to hex base is undefined. Square or radius on end of terminal is undefined.
5. A chamfer (or undercut) on one or both ends of hexagonal portions is optional.
6. Tabulation optional.
7. Minimum flat.
8. Flexible leads.

Letter	Dimensions				Notes
	Inches		Millimeters		
	Min	Max	Min	Max	
A		1.520		38.10	4
B	.530	.755	13.46	19.18	
C	.063	.172	1.60	4.37	
ϕD		1.100		27.94	
ϕD_1		.600		15.24	
E	1.218	1.252	30.94	31.75	
F	.250	.562	6.35	14.27	5
H	5.125	6.750	130.18	171.45	
I_1	.375		9.53		7
ϕM	.660	.745	16.76	18.92	2
M_1		.125		3.18	6
N	.793	.828	20.14	21.03	
Q		2.300		57.15	
Q_1		.375		9.53	6
ϕt	.265	.350	6.73	3.89	
W					3

FIGURE 1. Physical dimensions for semiconductor devices.

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PAGE 4

4.3, add the following screen to the screening table:

"

Screen (see table II of MIL-PRF-19500	Measurement
	JANTX and JANTXV levels
1	Method 2073 may be used in lieu of 2074 for compression bonded devices only.

"

PAGE 7

* TABLE I, subgroup 6, surge current, conditions column: Delete I_{FSM} = 6250 a(pk) and substitute I_{FSM} = 5000 a(pk) .

PAGE 8

* TABLE II, subgroup 2, Hermetic seal, Fine leak, delete in its entirety.

* TABLE II, subgroup 3, Blocking Life: Delete $T_C = 200^{\circ}C$ and substitute $T_C = 185^{\circ}C$

TABLE II, subgroup 5: Delete "$R_{\theta JC}$ = 0.15°C/W" and substitute "$R_{\theta JC}$ = 0.20°C/W".

* TABLE II, subgroup 7, power cycling: Delete T_C (high) = 150°C +5°C, -0°C and substitute T_C (high) = 135°C +5°C, -0°C

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*TABLE III, subgroup 3, delete in its entirety and substitute in the inspection column:

Subgroup 3
Not applicable

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* TABLE III, subgroup 5, barometric pressure (reduced), Inspection column, add For qualification only.

* TABLE III, subgroup 5, barometric pressure (reduced), delete:

1N3168, 1N3170		Pressure = 8 mm Hg	
1N3172, 1N3174		Pressure = 15 mm Hg	
1N3175, 1N3176		Pressure = 33 mm Hg	
1N3177		Pressure = 54 mm Hg	

and substitute:

1N3168		Pressure = 8 mm Hg	
1N3170		Pressure = 16 mm Hg	
1N3172		Pressure = 30 mm Hg	
1N3174		Pressure = 54 mm Hg	
1N3175		Pressure = 72 mm Hg	
1N3176		Pressure = 90 mm Hg	
1N3177		Pressure = 108 mm Hg	

* TABLE III, subgroup 6, blocking life, conditions column: Delete $T_C = 200°C$ and substitute $T_C = 185°C$.

The margin of this amendment is marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

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